MATH 180 Strategies of Problem Solving

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Outline

1. SoPS Lecture 14
   - Connect 4 Problems
   - Ready Set Bet
   - Wrapping up

Review

Last time

Review

Last time

- Connect 4

Review

Last time

- Connect 4
- weak solutions vs strong solutions

Review

Last time

- Connect 4
- weak solutions vs strong solutions
- visualizing a Connect 4 weak solution

Outline

1. SoPS Lecture 14
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   - Ready Set Bet
   - Wrapping up

Combinatorics problem

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Problem:

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Problem:

You have 10 identical marbles, a blue cup, and a red cup.

Combinatorics problem

Problem:

You have 10 identical marbles, a blue cup, and a red cup.
How many different ways can we put the marbles into the cups?

Combinatorics problem

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Solution:

Combinatorics problem

Solution:

- 10 in blue, 0 in red;

Combinatorics problem

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- 10 in blue, 0 in red;
- 9 in blue, 1 in red;

Combinatorics problem

Solution:

- 10 in blue, 0 in red;
- 9 in blue, 1 in red;
- 8 in blue, 2 in red;

Combinatorics problem

Solution:

- 10 in blue, 0 in red;
- 9 in blue, 1 in red;
- 8 in blue, 2 in red;
- . . .

Combinatorics problem

Solution:

- 10 in blue, 0 in red;
- 9 in blue, 1 in red;
- 8 in blue, 2 in red;
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- 1 in blue, 9 in red;

Combinatorics problem

Solution:

- 10 in blue, 0 in red;
- 9 in blue, 1 in red;
- 8 in blue, 2 in red;
- . . .
- 1 in blue, 9 in red;
- 0 in blue, 10 in red.

Combinatorics problem

Solution:

- 10 in blue, 0 in red;
- 9 in blue, 1 in red;
- 8 in blue, 2 in red;
- . . .
- 1 in blue, 9 in red;
- 0 in blue, 10 in red.

11 total ways

Combinatorics problem – version 2

Combinatorics problem – version 2

Problem:

Combinatorics problem – version 2

Problem:

You have 10 identical marbles, a blue cup, a red cup, and a yellow cup.

Combinatorics problem – version 2

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How many different ways can we put the marbles into the cups?

Combinatorics problem – version 2

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Solution:

Combinatorics problem – version 2

Solution:

- 10 in blue, 0 in red, 0 in yellow;

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- 10 in blue, 0 in red, 0 in yellow;
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- 10 in blue, 0 in red, 0 in yellow;
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Solution:

- 10 in blue, 0 in red, 0 in yellow;
- 9 in blue, 1 in red, 0 in yellow;
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- . . .
- 1 in blue, 9 in red, 0 in yellow;

Combinatorics problem – version 2

Solution:

- 10 in blue, 0 in red, 0 in yellow;
- 9 in blue, 1 in red, 0 in yellow;
- 8 in blue, 2 in red, 0 in yellow;
- . . .
- 1 in blue, 9 in red, 0 in yellow;
- 0 in blue, 10 in red, 0 in yellow;

Combinatorics problem – version 2

Solution:

- 10 in blue, 0 in red, 0 in yellow;
- 9 in blue, 1 in red, 0 in yellow;
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- 1 in blue, 9 in red, 0 in yellow;
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11 total ways

Combinatorics problem – version 2

Solution (continued):

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Solution (continued):

- 9 in blue, 0 in red, 1 in yellow;

Combinatorics problem – version 2

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- 9 in blue, 0 in red, 1 in yellow;
- 8 in blue, 1 in red, 1 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 9 in blue, 0 in red, 1 in yellow;
- 8 in blue, 1 in red, 1 in yellow;
- 7 in blue, 2 in red, 1 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 9 in blue, 0 in red, 1 in yellow;
- 8 in blue, 1 in red, 1 in yellow;
- 7 in blue, 2 in red, 1 in yellow;
- . . .

Combinatorics problem – version 2

Solution (continued):

- 9 in blue, 0 in red, 1 in yellow;
- 8 in blue, 1 in red, 1 in yellow;
- 7 in blue, 2 in red, 1 in yellow;
- . . .
- 1 in blue, 8 in red, 1 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 9 in blue, 0 in red, 1 in yellow;
- 8 in blue, 1 in red, 1 in yellow;
- 7 in blue, 2 in red, 1 in yellow;
- . . .
- 1 in blue, 8 in red, 1 in yellow;
- 0 in blue, 9 in red, 1 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 9 in blue, 0 in red, 1 in yellow;
- 8 in blue, 1 in red, 1 in yellow;
- 7 in blue, 2 in red, 1 in yellow;
- . . .
- 1 in blue, 8 in red, 1 in yellow;
- 0 in blue, 9 in red, 1 in yellow;

10 total ways

Combinatorics problem – version 2

Solution (continued):

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;
- 7 in blue, 1 in red, 2 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;
- 7 in blue, 1 in red, 2 in yellow;
- 6 in blue, 2 in red, 2 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;
- 7 in blue, 1 in red, 2 in yellow;
- 6 in blue, 2 in red, 2 in yellow;
- . . .

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;
- 7 in blue, 1 in red, 2 in yellow;
- 6 in blue, 2 in red, 2 in yellow;
- . . .
- 1 in blue, 7 in red, 2 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;
- 7 in blue, 1 in red, 2 in yellow;
- 6 in blue, 2 in red, 2 in yellow;
- . . .
- 1 in blue, 7 in red, 2 in yellow;
- 0 in blue, 8 in red, 2 in yellow;

Combinatorics problem – version 2

Solution (continued):

- 8 in blue, 0 in red, 2 in yellow;
- 7 in blue, 1 in red, 2 in yellow;
- 6 in blue, 2 in red, 2 in yellow;
- . . .
- 1 in blue, 7 in red, 2 in yellow;
- 0 in blue, 8 in red, 2 in yellow;

9 total ways

Combinatorics problem – version 2

Solution (continued):

Combinatorics problem – version 2

Solution (continued):
The pattern continues ... until:

Combinatorics problem – version 2

Solution (continued):
The pattern continues ... until:

- 0 in blue, 0 in red, 10 in yellow;

Combinatorics problem – version 2

Solution (continued):

Combinatorics problem – version 2

Solution (continued):
Now we can get a Grand Total: $11 + 10 + 9 + \cdots + 1$.

Combinatorics problem – version 2

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Remember Gauss's formula

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Combinatorics problem – version 2

Solution (continued):
Now we can get a Grand Total: $11 + 10 + 9 + \cdots + 1$.
Remember Gauss's formula

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Using this we get our final answer:

66 total ways

Combinatorics problem – version 3

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Problem:

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You have 10 identical marbles, a blue cup, a red cup, a yellow cup, and a green cup.

Combinatorics problem – version 3

Problem:

You have 10 identical marbles, a blue cup, a red cup, a yellow cup, and a green cup.
How many different ways can we put the marbles into the cups?

Combinatorics problem – version 3

Problem:

You have 10 identical marbles, a blue cup, a red cup, a yellow cup, and a green cup.
How many different ways can we put the marbles into the cups?

- we could approach it like before

Combinatorics problem – version 3

Problem:

You have 10 identical marbles, a blue cup, a red cup, a yellow cup, and a green cup.
How many different ways can we put the marbles into the cups?

- we could approach it like before
- might need *sum of squares formula*

Combinatorics problem – version 3

Problem:

You have 10 identical marbles, a blue cup, a red cup, a yellow cup, and a green cup.
How many different ways can we put the marbles into the cups?

- we could approach it like before
- might need *sum of squares formula*
- is there an easier way?

Combinatorics problem – general version

Problem:

You have n identical marbles and n different cups.
How many different ways can we put the marbles into the cups?

Combinatorics problem – general version

Problem:

You have n identical marbles and n different cups.
How many different ways can we put the marbles into the cups?

Solution:

We will use an argument called **stars and bars**.

Stars and bars

Stars and bars

- Label the cups 1 through n.

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- Marble arrangments are described by m stars with $n-1$ bars.

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- Label the cups 1 through n.
- Marble arrangments are described by m stars with $n-1$ bars.
- How many stars before the first bar is how many marbles in the first cup.
- How many stars after the first bar and before the second is how many marbles in the second cup.

Stars and bars

- Label the cups 1 through n.
- Marble arrangments are described by m stars with $n-1$ bars.
- How many stars before the first bar is how many marbles in the first cup.
- How many stars after the first bar and before the second is how many marbles in the second cup.
- Continue this pattern until, the number of stars after the last bar repesents the number of marbles in the n'th cup.

Stars and bars – example

$$\star\star\,|\,\star\,||\,\star\star\star\,|\,\star\star$$

Stars and bars – example

$$\star\star\,|\,\star\,||\,\star\star\star\,|\,\star\star$$

- 4 bars = 5 cups

Stars and bars – example

$$\star\star\,|\,\star\,||\,\star\star\star\,|\,\star\star$$

- 4 bars = 5 cups
- 8 stars = 8 marbles

Stars and bars – example

$$\star\star\,|\,\star\,||\,\star\star\star\,|\,\star\star$$

- 4 bars = 5 cups
- 8 stars = 8 marbles
- arrangement:
  - two marbles in cup 1;
  - one in cup 2;
  - zero in cup 3;
  - three in cup 4;
  - two in cup 5.

Challenge:

Problem:

Consider the following **stars and bars** expression:

$$||\star\star\star|\star\star|\star\star\star|$$

Challenge:

Problem:

Consider the following **stars and bars** expression:

$$||\star\star\star|\star\star|\star\star\star|$$

- how many cups in this arrangement

Challenge:

Problem:

Consider the following **stars and bars** expression:

$$||\star\star\star|\star\star|\star\star\star|$$

- how many cups in this arrangement
- how many marbles in this arrangement

Challenge:

Problem:

Consider the following **stars and bars** expression:

$$||\star\star\star|\star\star|\star\star\star|$$

- how many cups in this arrangement
- how many marbles in this arrangement
- describe how many marbles in each cup

Challenge:

Problem:

Suppose you have 5 cups with 3 marbles in the first cup 2 in the second, 4 in the third, 0 in the fourth and 1 in the fifth. Determine the stars and bars expression which describes this arrangement.

Stars and bars formula

Theorem (Stars and bars formula):

The number of ways of putting n indistinguishable marbles into m different cups is $\binom{m+n-1}{m}$.

Stars and bars formula

Theorem (Stars and bars formula):

The number of ways of putting n indistinguishable marbles into m different cups is $\binom{m+n-1}{m}$.

Proof:

It's the same as the number of stars and bars expressions with m stars and $n-1$ bars. Each expression is a string with $m+n-1$ characters, of which you choose m spots for stars and fill the rest with bars. Thus there's $m+n-1$ choose m total ways.

Stars and bars formula

Examples:

Stars and bars formula

Examples:

- with 10 marbles and two cups: $\binom{10+2-1}{10} = \binom{11}{10} = 11$

Stars and bars formula

Examples:

- with 10 marbles and two cups: $\binom{10+2-1}{10} = \binom{11}{10} = 11$
- with 10 marbles and three cups: $\binom{10+3-1}{10} = \binom{12}{10} = 66$

Stars and bars formula

Examples:

- with 10 marbles and two cups: $\binom{10+2-1}{10} = \binom{11}{10} = 11$
- with 10 marbles and three cups: $\binom{10+3-1}{10} = \binom{12}{10} = 66$
- with 10 marbles and four cups: $\binom{10+4-1}{10} = \binom{13}{10} = 286$

Stars and bars formula

Examples:

- with 10 marbles and two cups: $\binom{10+2-1}{10} = \binom{11}{10} = 11$
- with 10 marbles and three cups: $\binom{10+3-1}{10} = \binom{12}{10} = 66$
- with 10 marbles and four cups: $\binom{10+4-1}{10} = \binom{13}{10} = 286$
- with 15 marbles and seven cups: $\binom{15+7-1}{15} = \binom{21}{15} = 54264$

In-class homework:

Activity:

Work on current in-class problems.

Outline

1. SoPS Lecture 14
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Play the game

Activity

Play the game

Activity

- play a couple rounds of Ready Set Bet

Play the game

Activity

- play a couple rounds of Ready Set Bet
- what strategies seem to be the best?

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Final thoughts

Reflection

Final thoughts

Reflection

- Explain in your own words how an expression with stars and bars represents an arrangement of identical marbles in different containers.

Submit your responses by **Tonight!**